

IK 517: Philosophy of Science

Course Code	IK 517
Course Name	Philosophy of Science
L-T-P-C	3-0-0-3
Prerequisites	-
Distribution	Elective
Mutual Exclusion	-
One-time approval	59th BoA (Apr 2025)

Course Contents

- **Foundations of the Philosophy of Science** [4 Lectures]
 - Introduction to the Philosophy of Science
 - Historical Background and Key Concepts
 - The Role of Philosophy in Science
- **Laws, Explanation, and Prediction** [4 Lectures]
 - The Value of Laws in Science
 - Explanation and Prediction in Scientific Theories
 - Case Studies in Scientific Explanation
- **Induction and Probability** [4 Lectures]
 - Induction and Statistical Probability
 - Induction and Logical Probability
 - Challenges to Inductive Reasoning
- **Measurement and Quantitative Language** [4 Lectures]
 - The Experimental Method
 - Quantitative Concepts in Science
 - Extensive Magnitudes and Measurement
- **Space, Time, and Geometry** [4 Lectures]
 - Euclidean and Non-Euclidean Geometries
 - Space in Relativity Theory
 - Philosophical Implications of Space and Time
- **Causality and Determinism** [4 Lectures]
 - Causality in Science
 - Determinism and Free Will
 - Case Studies in Causality and Determinism
- **Theories and Nonobservables** [5 Lectures]
 - Theories and Correspondence Rules
 - The Ramsey Sentence
 - Analyticity in Scientific Theories
- **Statistical Laws and Quantum Mechanics** [5 Lectures]
 - Statistical Laws and Indeterminism
 - The Philosophy of Quantum Physics
 - Case Studies in Quantum Mechanics
- **Ethics, Technology, and Social Dimensions of Science** [4 Lectures]
 - Ethics in Science
 - The Role of Technology in Science
 - Social Dimensions of Science
- **Philosophy of Scientific Revolutions and Contemporary Issues** [4 Lectures]
 - Philosophy of Scientific Revolutions
 - Contemporary Issues in Philosophy of Science
 - Review and Synthesis

Text and Reference Books

1. Rudolf Carnap, **An Introduction to the Philosophy of Science**, Primary Text.
2. James Ladyman, **Understanding Philosophy of Science**, Supplementary Text.